

PumpTrack Test Plan Document Information Field Details Project Name PumpTrack – Fuel Station Finder & Management System Version 1.0 Prepared By Smitha Shetty College CSMIT College, Panvel Project Guide Santham Bharathy Date (Enter today's date) Document Type Test Plan

1. Introduction

PumpTrack is a web-based Fuel Station Finder and Management System developed using React.js and Firebase. The application enables users to register, log in, search nearby fuel stations, check fuel availability, view queue status, access Google Maps navigation, and manage their profile. The application also provides an Admin module to manage fuel station information.

This Test Plan defines the testing strategy, objectives, scope, resources, deliverables, and schedule for validating the PumpTrack application.

2. Objectives

The objectives of testing are:

Verify that all functional requirements are implemented correctly. Ensure the application performs as expected. Identify defects before deployment. Validate user authentication and authorization. Verify CRUD operations in the Admin module. Ensure search and filtering functions work correctly. Verify map integration and navigation. Ensure compatibility across supported browsers. Deliver a stable and reliable application.

3. Project Description

PumpTrack provides users with a convenient way to locate nearby fuel stations and check their availability.

The system includes:

User Registration User Login Dashboard Station Search Station Details Google Maps Navigation User Profile Admin Dashboard Fuel Station Management CSV Export

4. Scope of Testing

In Scope

The following modules will be tested:

User Registration User Login Logout Dashboard Search Functionality Station Details Fuel Availability Queue Status Profile Management Admin Dashboard Add Station Edit Station Delete Station CSV Export Google Maps Navigation

Out of Scope

The following are not included in this project:

Payment Gateway Fuel Booking Online Transactions SMS Notifications Push Notifications

5. Modules to be Tested

Module ID	Module Name
M001	Registration
M002	Login
M003	Dashboard
M004	Search
M005	Station Details
M006	Profile
M007	Map
M008	Admin Dashboard
M009	CRUD Operations
M010	CSV Export

6. Testing Types

The following testing types will be performed:

Smoke Testing Sanity Testing Functional Testing Integration Testing System Testing Regression Testing User Acceptance Testing (UAT) Compatibility Testing Responsive Testing Usability Testing

7. Test Environment

Component	Details
Operating System	Windows 11
Browser	Google Chrome
Frontend	React.js
Backend	Firebase Database Firestore
IDE	Visual Studio Code
Deployment	Vercel
Version Control	GitHub

8. Entry Criteria

Testing will begin when:

Development is completed. Build is deployed successfully. Firebase is configured. Test data is available. Application is accessible. 9. Exit Criteria

Testing will be completed when:

All planned test cases are executed. Critical defects are resolved. Smoke testing passes. Functional testing passes. Regression testing passes. UAT is approved. 10. Test Deliverables

The following documents will be prepared:

Test Plan Test Scenarios Test Cases Requirement Traceability Matrix (RTM) Bug Report Test Execution Report Smoke Test Report Sanity Test Report Functional Test Report Integration Test Report System Test Report Regression Test Report UAT Report Test Summary Report 11. Risks

Possible risks include:

Internet connectivity issues. Firebase service downtime. Browser compatibility issues. Invalid test data. Unauthorized access attempts. Changes in project requirements. 12. Test Schedule Activity Status Test Planning Completed Test Scenario Preparation Pending Test Case Design Pending Test Execution Pending Defect Reporting Pending Regression Testing Pending UAT Pending Test Summary Report Pending 13. Roles and Responsibilities Role Responsibility Developer Develop and fix defects Tester Execute test cases and report defects Project Guide Review and approve project Student Prepare documentation and perform testing 14. Approval Name Role Smitha Shetty Developer & Tester Santham Bharathy Project Guide Document Status

Version: 1.0

Status: Approved for Testing